Curriculum vitae: Gaia Grosso

Personal Data

Name:	Gaia Grosso
Nationality:	Italian
Date and Place of Birth:	May 7, 1995 - Treviso (Italy)
email (1):	gaiag795@mit.edu
email (2):	gaia.grosso@cern.ch
Languages:	Italian (mother tongue), English (fluent), French (basic), Catalan (intermediate)

Work Experience

Sep. 2023 – Sep. 2026 | IAIFI postdoctoral fellow

NSF AI Institute for Artificial Intelligence and Fundamental Interactions (IAIFI) – Cambridge, MA, US

Short description: As an IAIFI fellow my research activity focuses on the application of cutting-edge AI techniques to particle physics data and the design of physics inspired AI algorithms. In collaboration with Prof. Harris' group at MIT, I work on the design of novel machine-learning-based statistical tools for signal-agnostic new physics discovery at collider and gravitational waves experiments. In collaboration with Prof. Ba's group at Harvard SEAS, I study physics inspired interpretable models for anomaly detection, data validation, and safety/alignment of generative AI models.

Mentors:
Prof. Philip Harris (Laboratory for Nuclear Science, Massachusetts Institute of Technology)
Prof. Demba Ba (School of Engineering and Applied Science, Harvard University)

Nov. 2019 – Jan. 2023 | CERN doctoral student

European Organization for Nuclear Research (CERN) – Geneva, Switzerland

Short description: As CERN doctoral student, I worked on machine-learning solutions for data analysis for the CMS experiment. The main activities involved: (1) the design of a framework to deal with systematic uncertainties in a machine-learning based strategy for agnostic New Physics searches; (2) Exploring models based on Graph Neural Networks for fast jets reconstruction on GPUs; (3) The development of a monitoring and control software interface for the acquisition boards deployed in the 40 MHz scouting at the L1 trigger of the CMS experiment.

Supervisor: Dr. Maurizio Pierini (CERN)

Jun. 2018 – Sep. 2018 | CERN Summer student

European Organization for Nuclear Research (CERN) – Geneva, Switzerland

Project: "Machine Learning techniques to search for New Physics"
Short description: I studied the machine-learning-based algorithm proposed by d'Agnolo & Wulzer for signal-agnostic new physics searches and I explored possible ways to successfully apply it to multivariate problems. Main results of the research activity include: (1) the analysis of the algorithm stability for some basic univariate cases; (2) the design of a regularization strategy for the application of the algorithm to multivariate scenarios; (3) preliminary tests on simulated LHC collision events with two muons in the final state.

Supervisors: Dr. Maurizio Pierini (CERN), Dr. Olmo Cerri (Caltech)

Report: https://cds.cern.ch/record/2653342

Education

Oct. 2019 – May 2023 — PhD in Physics
University of Padova (Italy)

Title: "Searching for Unexpected New Physics at the LHC with Machine Learning"

PhD courses:
Quantum tools for future scientific research, Future Colliders,
Statistics for data analysis, Physics models for data analysis,
EU funding: opportunities for Research and Innovation and proposal writing

Thesis: "Searching for Unexpected New Physics at the LHC with Machine Learning" [link]

Short description: I developed the New Physics Learning Machine (NPLM), a machine learning–based, model-agnostic strategy to test for data departures from the Standard Model at the LHC. The method approximates a maximum-likelihood hypothesis test by using a neural network to model the data distribution and compute a likelihood-ratio, returning a p-value for compatibility with the Standard Model. I devised a strategy to include reference-model uncertainties and deployed the algorithm in practical settings: applying it to a real LHC new-physics search in the dimuon channel (CMS, 13 TeV) and to data-quality monitoring of drift tube chambers. The approach supports high-dimensional problems, identifies the most discrepant phase-space regions, and reconstructs multidimensional distributions for interpretability.

Publication: This work resulted in several journal publications: Eur. Phys. J. C 82 (2022) 3, 275); Eur.Phys.J.C 82 (2022) 10, 879 ; Mach.Learn.Sci.Tech. 4 (2023) 3, 035029; SciPost Phys. 16 (2024) 123.

Supervisors: Prof. Marco Zanetti, Prof. Andrea Wulzer

Oct. 2017 – Sep. 2019 — Master's degree in Physics
University of Padova (Italy)

Grade: 110/110, cum laude

Curriculum: Physics of fundamental interactions

Optional courses: Stochastic methods, Statistical learning, Big data computing

Thesis: "Unbiased ML-based strategy for New Physics searchs at the LHC"
Short description: Designed and implemented a neural-network–driven hypothesis test for model-agnostic anomaly detection at the LHC, approximating the Neyman–Pearson optimal statistic. Extended the method from the univariate toy setting to realistic multivariate analyses, introducing a robust tuning strategy and showing performance gains with luminosity, laying groundwork for future inclusion of systematics and large-scale computing.

Publication: This work resulted in a journal publication Eur. Phys. J. C 81 (2021) 1, 89

Supervisor: Prof. Marco Zanetti

Oct. 2014 – Sep. 2017 — Bachelor's degree in Physics
University of Padova (Italy)

Grade: 106/110
Thesis: "Deep Learning techniques to search for New Physics at the LHC" [link]

Short description: I applied deep learning to distinguish signal from background in LHC collision data, training neural networks directly on raw kinematic features. I showed that deep architectures can equal or surpass shallow networks trained on engineered non-linear variables, offering a powerful, flexible analysis tool for new-physics searches.

Supervisor: Prof. Marco Zanetti

Selected Publications

full record at: ORCID 0000-0002-8303-3291

- **Grosso G.**, Hindupur S.S.R., Fel T., Bright-Thonney S., Harris P., Ba D. Sparse, self-organizing ensembles of local kernels discover rare statistical anomalies *arXiv-preprint* [2511.03095] submitted to PNAS
- Cappelli P., **Grosso G.**, Letizia M., Reyes-Gonzalez H., Zanetti M. Learning to Validate Generative Models: a Goodness-of-Fit Approach *arXiv-preprint* [2511.09118] submitted to Mach. Learn. Sci. Tech.
- Bright-Thonney S., Reissel C., **Grosso G.**, Woodward N., Govorkova E., Novak A, Park S.E., Moreno E., Harris P AutoSciDACT: Automated Scientific Discovery through Contrastive Embedding and Hypothesis Testing *accepted at NeurIPS 2025* [2510.21935]
- Metzger K., Sodini M., Xu L., Aarrestad T., Govorkova E., **Grosso G.**, Harris P. Anomaly preserving contrastive neural embeddings for end-to-end model-independent searches at the LHC. *Phys. Rev. D 112, 072011* [DOI]
- **Grosso G.**, Sengupta D., Golling T., Harris P. Robust resonant anomaly detection with NPLM. *Eur. Phys. J. C 85 9 (2025)* [DOI]
- **Grosso G.**, Letizia M. Multiple testing for signal-agnostic searches of new physics with machine learning. *Eur. Phys. J. C 85, 4 (2025)* [DOI]
- **Grosso G.** Anomaly-aware summary statistic from data batches. *J. High Energ. Phys. 2024, 93 (2024)* [DOI]
- **Grosso G.**, Letizia M., Wulzer A., Pierini M. Goodness of fit by Neyman-Pearson testing. *SciPost Phys. 16, 123 (2024)* [DOI]
- **Grosso G.**, Lai N., Letizia M., Pazzini J., Rando M., Rosasco L., Wulzer A., Zanetti M. Fast kernel methods for Data Quality Monitoring as a goodness-of-fit test.
  Mach. Learn. Sci. Tech. 4 (2023) 3, 035029 [DOI]
- Letizia M., Losapio G., Rando M., **Grosso G.**, Wulzer A., Pierini M., Zanetti M., Rosasco L. Learning new physics efficiently with nonparametric methods. *Eur. Phys. J. C* 82(10):879 (2022). [DOI]
- D'Agnolo R.T., **Grosso G.**, Pierini M., Wulzer A., Zanetti M., Learning new physics from an imperfect machine. *Eur. Phys. J. C* 82(3):275, (2022). [DOI]
- D'Agnolo R.T., **Grosso G.**, Pierini M., Wulzer A., Zanetti M., Learning Multivariate New Physics. *Eur. Phys. J. C* 81(1):89 (2021). [DOI]

Conferences and Workshops

Jul. 2026 | Mechanistic Interpretability Workshop
ICML (Seoul, SK)
Poster (main-author): Adaptive and interpretable two-sample test as a batch-level tool for AI safety [link]

Jun. 2026 | 2026 Conference on Physics and AI (PAI26)
Stanford, CA, US
Poster (co-author): Covariant Contrastive Learning for Uncertainty-Aware Anomaly Detection [link]

Apr. 2026 | Statistical Physics for Ambitious Interpretability
Cambridge, MA, US
Invited Speaker: Detecting statistical anomalies with sparse, self-organizing ensembles of local kernels [link]

Dec. 2025 | 8th Workshop on Machine Learning and the Physical Sciences
NeurIPS (San Diego, CA, US)
Poster (co-author): An end-to-end pipeline for uncertainty-aware validation of generative AI [link]

The Thirty-Ninth Annual Conference on Neural Information Processing Systems
NeurIPS (San Diego, CA, US)
Poster (co-author): AutoSciDACT: Automated Scientific Discovery through Contrastive Embedding and Hypothesis Testing [link]

Oct. 2025 | STAMPS-NSBI Workshop
Carnegie-Mellon University (Pittsburgh, PA, US)
Invited talk: From high dimensions to statistical discovery: anomaly detection at scale in the era of latent spaces [YouTube]

Sep. 2025 | FirstPrinciple Talks Program
remote
Talk: Validation, monitoring and discovery: statistical tests for anomalies in the era of latent spaces

Building Big or Building Smart, MIAPbP workshop
TUM (Munich, Germany)
Talk: Anomaly Detection in the era of Latent Spaces. [link]

Jun. 2025 | Anomaly Detection for HEP workshop
Columbia University, (Irvington, NY, US)
Talk: From High Dimensions to Statistical Discovery: A Contrastive Learning Approach to Anomaly Detection [link]

European AI for Fundamental Physics Conference (EuCAIFCon)
(Cagliari, Italy)
Poster (co-author): End-to-End Optimization of Generative AI for Robust Background Estimation [link]

Mar 2025 | PHYSTAT informal review
remote
Round table: GOF tests via Machine Learning [link]

Dec. 2024 | 7th Workshop on Machine Learning and the Physical Sciences
NeurIPS (Vancouver, BC, CA)
Poster (main author): Learning dictionaries of New Physics with sparse local kernels [link]
Poster (co-author): Product Manifold Machine Learning for Physics [link]

Sep. 2024 | PHYSTAT - "Statistics meets ML" in Particle Physics & Astrophysics
Imperial College (London, UK)
Invited talk: Statistical tests for anomaly detection at the LHC. [link]

Aug. 2024 | IAIFI workshop
MIT (Cambridge, MA, US)
Talk: How good is your model? — Goodness-of-fit by Neyman-Pearson testing. [link]

May 2024 | 22nd International Workshop on Advanced Computing and Analysis Techniques in Physics Research (ACAT24)
Stony Brook University (Long Island, NY, US)
Talk: Boosting statistical anomaly detection via multiple test with NPLM. [link]

Mar. 2024 | European AI for Fundamental Physics Conference (EuCAIFCon)
(Amsterdam, NE)
Poster (main author): Building sparse kernel methods via dictionary learning. Expressive, regularized and interpretable models for statistical anomaly detection [link]

Nov. 2023 | From Physics to Medicine: XAI workshop
Invited talk: Uncertainty aware ML models [link]

Sep. 2023 | Institute of Advanced Study 2023
University of Geneva (Geneva, Switzerland)
Public talk: How good is the Standard Model? Hunting anomalies in the LHC data and beyond [link]

Jun. 2023 | PHYSTAT-2sample 2023
(remote)
Talk: Goodness of fit by Neyman-Pearson testing [link]

May 2023 | 26th International conference on Computing in High Energy Physics (CHEP 2023)
Norfolk (VA, US)
Talk: Systematics in ML model independent searches [link]

Apr. 2023 | Systematic Effects and Nuisance Parameters in Particle Physics Data Analyses
Banff International Research Station (Banff, Canada)
Talk: Unbiased detection of data departures from expectations with machine learning [link]

Dec. 2022 | 5th Workshop on Machine Learning and the Physical Sciences
NeurIPS (Vancouver, BC, CA)
Poster (main author): How good is the Standard Model? Machine learning multivariate Goodness of Fit tests [link]
Poster (co-author): A fast and flexible machine learning approach to data quality monitoring [link]

Sep. 2022 | Machine Learning at GGI
Galielo Galilei Institute (Firenze, Italy)
Seminar: New Physics Learning Machine (NPLM) at work [link]
Conference talk: Learning New Physics from a Machine [link]

Aug. 2022 | Hammers&Nails
Weizmann Institute of Science (Rehovot, Isreal)
Talk: New Physics Learning Machine (NPLM) at work [link]

May 2022 | 5th inter-experiment Machine Learning workshop
CERN (Geneva, Switzerland)
Talk: Learning New Physics aware of systematic uncertainties [link]

Apr. 2022 | Learning to Discover
Institut Pascal (Paris-Saclay, France)
Talk: An Imperfect Machine to search for New Physics: dealing with uncertainties in a machine-learning based signal extraction [link]

Spring workshop on Physics of Data
Istituto Veneto di Scienze Lettere ed Arti (Venice, Italy)
Talk: Model-independent search for New Physics at the LHC [link]

Dec., 2021 | 4th Workshop on Machine Learning and the Physical Sciences
NeurIPS (remote participation)
Poster: An Imperfect machine to search for New Physics: systematic uncertainties in a machine-learning based signal extraction [link]

20th International Workshop on Advanced Computing and Analysis Techniques in Physics Research (ACAT21)
(remote)
Talk: An Imperfect Machine to search for New Physics: a way to include systematic uncertainties [link]

Teaching

Mar. 30, 2026 | AI Methods for Science, CDS DS 595, Spring 2026
Boston University (Boston, MA)

Guest Lecture: "Same or Different? Classical and ML methods for two-sample testing in Science"
Short description: Guest lecturer on two-sample hypothesis testing methods for scientific discovery, covering classical statistics, kernel MMD, and classifier-based tests with applications to high-energy physics and AI model validation.

Nov. 17–21, 2025 | First PHYSTAT School of Statistics: "Statistics in the era of Machine Learning"
Nikhef (Nooitgedacht, NE)

Lecture: "Statistical and ML tools for anomaly detection in fundamental physics"
Short description: In this lecture I survey statistical and machine-learning tools for anomaly detection in fundamental physics, covering both classical hypothesis-testing techniques and modern approaches such as density estimation, representation learning, and weak supervision. Emphasis is placed on robustness, calibration, and uncertainty quantification, highlighting how methodological design choices impact sensitivity, reliability, and scientific interpretability.

Aug. 3–7, 2025 | 2025 IAIFI summer school
Harvard University (Cambridge, MA, US)

Lecture: "Physics-Guided Optimizations: Inform and Inspire AI Models with Physics Principles".
Short description: In this lecture I present key concepts and techniques for designing machine learning models both informed by and inspired by physics principles. In the first part, we explore ways to inform AI models about physics constraints and data properties. We examine the highly structured nature of data in the physical sciences —such as images, point clouds, and time series—and discuss how their geometric and physical properties shape effective machine learning approaches. We then cover practical strategies for incorporating physical constraints through loss functions, regularization, and inductive biases. In the second part, we highlight how physical principles can inspire the design of optimization methods and AI architectures, introducing concepts like annealing, diffusion, entropy, and energy-based models, and showing examples of their growing role in modern machine learning.

April 14, 2025 | MIT 8.16/8.316: Data Science in Physics
Massachusetts Institute of Technology (Cambridge, MA, US)

Guest lecture on the topic: "Statistical tests for anomaly detection and scientific discovery"
Short description: In this lecture I review statistical and machine-learning approaches to anomaly detection in fundamental physics, from classical hypothesis-testing frameworks to contemporary methods based on density estimation, representation learning, and weak supervision. Particular attention is given to robustness, calibration, and uncertainty quantification.

Aug. 5–9, 2024 | 2024 IAIFI summer school
Massachusetts Institute of Technology (Cambridge, MA, US)

Facilitator on the topic: "Diffusion models for physics".
Short description: I prepared instructional exercises covering (1) diffusion models, (2) variational autoencoders and (3) validation methods available as jupyter notebook on GitHub]. During the hands-on session, I guided the students through the problems and moderated interactive discussions among them.

Mentoring

Research advisor for graduate student at MIT

Apr. 2025 – ongoing | Shelley Tong

Title: Modeling continuous systematic uncertainties in contrastive spaces for LHC data
Short description: We explore how to incorporate continuous systematic uncertainties into contrastive-learning–based representations for anomaly detection for scientific discovery. Building on recent work showing that physics-informed contrastive learning yields structured latent spaces, the project focuses on extending these embeddings to incorporate continuous theoretical and experimental uncertainties. We use a downstream parameterized classification task to force the latent space to covary predictably with nuisance-parameter distortions, enabling transparent and controllable uncertainty propagation. Using simulated data from collider experiments, the aim is to demonstrate that this strategy produces low-dimensional embeddings that remain robust under systematic shifts, providing a reliable foundation for downstream model-agnostic anomaly detection.

Supervision of undergraduate student at Harvard University

Oct. 2025 – ongoing | Matilda Genieser

Title: Detect AI-generated or hallucinated text with SparKer
Short description: Matilda is building a demo to detect and visualize anomalous behaviors in text data using a new anomaly detection algorithm that I led the development of. Matilda is learning the basics of PyTorch and tensor manipulation by implementing a simple 2D prototype and visualizing its behavior. She will then scale the approach to real text data using text embeddings, building a research-level demonstration of how to detect anomalous text inAI-generated content.

Co-supervision of graduate student at Harvard University

Sep. 2025 – ongoing | Cameron McNamee

Title: Robust and interpretable local layers for supervised tasks.
Short description: Cam is studying the properties of machine-learning architectures based on local activations for supervised tasks such as classification. Together with another postdoc, I am advising him on developing a physics-inspired interpretation of the model and characterizing its robustness from a statistical perspective.

Research advisor of graduate student at Harvard University

Aug. 2025 – Oct. 2025 | Sai Sumedh R. Hindupur

Title: Learning dynamics of SparKer
Short description: I advised Sumedh in developing the theoretical description of SparKer — a new architecture I designed for anomaly detection in learned representations. SparKer integrates sparsity, locality, and competitive self-organization to produce geometrically interpretable detection experts. Together, we analyzed the model's learning dynamics, derived formal theoretical guarantees, and established rigorous proofs underpinning its statistical reliability.
Publications: Sumedh's work contributed to the paper pre-print arXiv:2511.03095, submitted to PNAS.

Research advisor of graduate student at the Institute for Theoretical Physics, IFT (Madrid, Spain)

May. 2025 – ongoing | Sergio Rodriguez Benitez

Title: Mixture of experts for weakly supervised anomaly detection
Short description: For new-physics discovery at collider experiments, Sergio is investigating anomaly-detection strategies that use binary classification coupled with a mixture-of-experts architecture, enabling the model to identify and leverage correlated features tailored to particular classes of signal hypotheses.

Research advisor for graduate student at University of Padova (Italy)

Sep. 2023 – ongoing | Nicoló Lai

Title: Signal-agnostic search for New Physics in the dimuon final state in proton-proton collision at $\sqrt{s} = 13$ TeV with the NPLM method (analysis internal to the CMS collaboration).
Short description: We are applying the NPLM methods for signal-agnostic new physics discovery to data collected by the CMS experiments. We analyze proton-proton collision with two opposite charged muons in the final state. The analysis rely on methodology developed during my PhD.
Publication: Work in progress internally to the CMS collaboration and not public yet. Public results are expected by September 2026.

Sep. 2023 – Sep. 2025 | Pietro Cappelli

Title: Generative model validation via machine-learning goodness-of-fit test
Short description: Pietro applied the New Physics Learning Machine (NPLM) to rigorously test and validate generative networks in high-dimensional scientific settings. The work demonstrated the power of the method on both synthetic Gaussian mixtures and a LHC jet generator, showing that NPLM not only detects deviations but also diagnoses regions where the generative model underperforms.
Publication: This work resulted in the paper pre-print arXiv:2511.09118 now under consideration in Mach. Learn. Sci. Tech.

Supervision of UROP undergraduate student at MIT

Oct. 2023 – Apr. 2025 | Lana Xu

Title: Latent representations for Anomaly Detection in particle physics.
Short description: Lana studied methods for neural embedding based on contrastive learning to build efficient representations for anomaly detection. She explored solutions based on MLP and Transformers using the VICReg loss as an objective.
Publication: Her work contributed to a journal publication Phys.Rev.D 112 (2025) 7, 072011.

Apr. 2023 – Apr. 2025 | Nathaniel S. Woodwards.

Title: Product Manifold Machine Learning for Physics
Short description: Nate developed and evaluated machine-learning architectures operating on product manifold (PM) representations (Cartesian products of constant-curvature Riemannian manifolds) for particle-physics data. Results showed that geometric tailoring the representation improves performance—especially for smaller models and highly hierarchical inputs—pointing.
Publication: this work was presented as a poster at ML4PS workshop at NeurIPS 2024 and resulted in the paper pre-print arXiv:2412.07033.

Co-supervision of ETHZ Master's students

Jan. 2024 – Jun. 2024 | Jessica Prendi

Title: Foraging New Physics: Model-agnostic new physics searches in the 40 MHz Scouting system of the CMS detector
Short description: This project studied a model-agnostic search strategy for new physics in the 40 MHz scouting data of the CMS experiment by combining tree-based anomaly detection with the New Physics Learning Machine (NPLM). We evaluated 11 anomaly-detection methods across 8 BSM benchmarks, identifying Isolation Forest as the most effective and competitive with variational autoencoders. We demonstrated that integrating NPLM with anomaly scores enables hypothesis testing without specifying a signal model, allowing reliable detection of small injected signals in dijet and MET spectra despite limited detector resolution. We showed that this combined approach enhances sensitivity, reduces trigger-induced bias, and provides a fully model-agnostic pipeline for real-time new-physics searches.

Oct. 2023 – Sep. 2024 | Kyle Sidney Metzger

Title: Representation learning for New Physics searches in the CMS experiment
Short description: The project applied neural representation learning to high-energy particle physics, comparing MLP and Transformer architectures under supervised and self-supervised objectives (SimCLR, SupCon, DINO). Learned embeddings significantly improved model-agnostic anomaly detection and open-world jet classification, supported by a new augmentation strategy tailored to particle-physics data. Latent-space analyses showed that jet-level features and aspects of jet substructure can emerge directly from particle-level inputs, highlighting the potential of scalable representation learning as a foundation for future particle-physics models.
Publication: This work resulted in a journal publication Phys.Rev.D 112 (2025) 7, 072011.

Nov. 2024 – Sep. 2025 | Giada Badaracco

Title: An end-to-end pipeline for uncertainty-aware validation of generative AI
Short description: Density estimation with generative Artificial Intelligence (AI) is a common task in the physical sciences, with applications ranging from particle physics to gravitational-wave parameter estimation. Many of the existing methods, however, do not provide a way to estimate epistemic uncertainties, which are essential for reliable hypothesis testing. We propose an end-to-end framework combining generative modeling with principled uncertainty quantification. A normalizing-flow ensemble is trained to synthesize events; ensemble-based epistemic uncertainties are computed and propagated into a learned likelihood–ratio goodness-of-fit (GoF) test. This yields robust distributional estimates that allow to synthesize significantly more events than those in the original training dataset and enable uncertainty-aware scientific discovery.
Publication: This work was accepted for a poster presentation at EuCaiF2025, and will be presented as a poster at the ML4PS workshop at NeurIPS (Dec 6, 2025). Full paper in preparation.

Management and organization

Apr. 2025 –ongoing | co-organizer of the VERAIPHY initiative

(Validation & Evaluation for Robust AI in Physics)
Role: I co-organize a collaborative effort endorsed by the PHYSTAT community aiming to gather a series of essays on the interplay between AI/ML methodologies and fundamental physics, with a strong focus on validation and evaluation procedures that enable statistically robust applications of these tools in scientific discovery.
Workshop: [Feb. 16-17, 2026]
Articles: [webpage]

Nov. 2025 –ongoing | CHEP 2026 track convener

planned for May 2026 (Chulalongkorn University in Bangkok, Thailand)
Role: I am serving as a convener for the track "Software environment and maintainability" at the 28th Conference on Computing in High Energy and Nuclear Physics. Main tasks include reviewing and selecting submitted abstracts; convening sessions; reviewing submitted proceedings after the conference.

Apr. 2023 –ongoing

CMS ML Innovation L3 convener

ML group, CMS experiment (CERN, Geneva, Switzerland)

Role: I organize seminars, workshops and hackathons on new machine learning topics relevant to particle physics problems and the application to data from the CMS experiment.

CMS ML hackathons I co-organized (internal):
- ML for reweighting (Oct. 23–25, 2023)
- Anomaly Detection for monitoring, validation and discovery (Jul. 1–3, 2024)

CMS ML workshops I co-organized (public):
- Foundation Models for Physics mini-workshop (Oct. 1–2, 2024) [link]
- Bites of Foundation Models for Physics (1): physics-inspired representations (Nov. 20, 2024) [link]
- Bites of Foundation Models for Physics (2): LLMs for experiments in fundamenta physics (June 3, 2025) [link]

Sep. 2024 – ongoing

IAIFI speaker selection committee

IAIFI (Cambridge, MA, US)

Role: Co-chairing the committee in charge of seminars and colloquia organization for IAIFI in the academic year 2024-25.

Mar. 14, 2025

IAIFI Research Panel on the 2024 Nobel Prize in Physics

Role: Co-organizer of the event

Short description: Given the 2024 Nobel Prize in Physics, awarded to John Hopfield and Geoffrey Hinton for their development of foundational and physics-inspired machine learning frameworks (the Hopfield network and the Bolzmann machine, respectively), we hosted a discussion explaining the research underlying the award and its widespread impact. The discussion included historical context and current developments in using physics for AI innovation. Experts at the intersection of physics, neuroscience, and computer science shared their perspectives on the research behind the Nobel Prize in Physics and the next great discoveries to come from similarly interdisciplinary, curiosity-driven research. [YouTube]

Nov. 1, 2024

IAIFI Fireside Chat on 2024 Physics Nobel Prize

Role: Co-organizer of the event

Short description: The 2024 Nobel Prize in Physics recognized John Hopfield and Geoffrey Hinton for their development of foundational and physics-inspired machine learning frameworks (the Hopfield network and the Boltzmann machine, respectively). Following on a fireside chat discussing the historical significance of the award, we hosted a panel discussion that dived deeper into the impact of Hopfield and Hinton's work, and more broadly of physics insights, on interdisciplinary research tackling the understanding of artificial and biological intelligent systems. [YouTube]

Mar 14–15, 2024

Symposium on the Impact of Generative AI in the Physical Sciences

MIT (Cambridge, MA, US)

Role: Co-organizer of the event

Short description: The IAIFI Symposium on the Impact of Generative AI in the Physical Sciences brought leading researchers in generative AI and physics together for plenary talks, panels, and discussion, to define the potential for this class of tools to impact physics research, and to understand what is needed for this potential to be realized. As part of the organizers committee I contributed to generate a whitepaper on the workshop topic.

Whitepaper: A Virtuous Cycle: Generative AI and Discovery in the Physical Sciences. Grosso G., Harris P., Mishra-Sharma S., and Shanahan P. *An MIT Exploration of Generative AI, September 2024.*

Reviewer

I actively serve as a reviewer for:

since 2025
- Physics Review D
- Nuclear Physics, Section B

since 2023
- Journal of High Energy Physics
- European Physics Journal C

since 2022
- Machine Learning for the physical sciences (NeurIPS workshop)

Outreach

Nov. 2024 | **Cambridge Science Festival** [web-page]
Cambridge (MA, US)
Outreach activity: I developed an LLM-powered interactive Marie Curie persona using retrieval-augmented generation; visitors could type in questions and read historically grounded responses synthesized in Curie's voice.

Museum of Science (Boston, MA)
Outreach talk: I gave a short presentation about AI and particle physics, introducing my research and the role of artificial intelligence to extend human capabilities. After the presentation I took part in an interactive Q&A.

Spring 2023 | **Science Gateway: "Physics at the LHC"**
CERN (Switzerland)
video recording for the exhibitions of the new Science Gateway center at CERN.

Nov. 2020 | **Veneto Night 2020:"the world of elementary particles for all ages"**
(virtual event)
Department of Physics and Astronomy "G. Galilei", University of Padova (Italy)
Livestream of a short presentation on New Physics searches at the LHC for general public, especially children [youtube].

2018 | **Cloud chamber for high school students**
Department of Physics and Astronomy "G. Galilei", University of Padova (Italy)
During my Master's studies I took part in a student research project aiming to build a cloud chamber. We use the chamber for outreach, to explain the main concepts of particle physics detection.